Ch=02 (Design Process)

* Human Computer Interaction:-

→ HCI is study & planned of HCI

→ HCI easy, safe, effective & enjoyable.

→ Goal of HCI to produce usable & safe system

- → Understand the factors
- → Develop tool & technique
- → Achieve efficient, effective & safe interact

* How a person interact with computer.

1) Identifies a task to be performed or need to be fullfilled.

2) Decide how the task will be completed or the need fulfilled.

3) Manipulates the computer's controls!.
   ↳ Keyboard, mouse → use to select choices, in list

4) Gather the necessary data!.

5) Forms judgments resulting in decisions relevant to the task or need.
   ↳ Data Valid? → Has the transaction Successfull.

* Factors that make system difficult to use

1) Use of Jargon :-
   - → words unknown to home environment.
   - → Learning to use system req learning a new lang.

2) Non-obvious design :-
   - → complex design are not obvious
   - → Result always related to action.

3) Fine distinction :-
   - → Different action may result same thing.
   - → Very little difference

4) Design incosiptency :-
   - → Same action may different names
   - → ex :- "Save", → keep to "write → list".

* Psychological response to poor design :-

1) confusion :- framework cannot be understood

2) Annoyance :- Inconsiptencies design,
   → slow computer reaction.

3) Frustration :- Overbundance of annoyance
   → Inability to finish a task.

4) Panic or stress :- , 5) Boredom.

* Physical reaction to poor design :-

1) Abondonment :- System is rejected & other info. source use.

2) Partial use of the system :- Only a portion of the system used.

3) Indirect use of the system :- An intermediary is placed between the user & computer.

4) Modication of the task :- The task is changed to match the capability of the system

5) Direct Programming :-
- → system is reprogrammed by user.
- → Meet specific need.

* Important human characteristic are :-

→ Perception = is our understanding through the physical sensation. sight, sound & smell.

1) Proximity

2) Similarity

3) Matching Pattern

4) Unity :- objective that form close shapes ~~are~~ perceived as a group

→ Memory:- → Short-term memory a limited amount of info.

→ Long-term memory contains the knowledge we process.

→ Sensory storage:- Is a buffer where automatic processing of info. collected from our senses.

→ It acts like radar, constantly scanning the environment

→ Visual acuity:- The capacity of the eye to resolve details is called visual activity.

→ Foveal & Peripheral vision:-

→ foveal vision used to focus directly on something

→ Peripheral vision senses anything in the surrounding

* Describe the following:-

a) Information Processing:-

1) Consciousness (higher level) :- It is limited
→ It is slow

2) Unconsciousness (lower level) :- → Rapidly
→ parallel with higher level without conscious effort.

→ Both function simultaneously H.L → reasoning
→ Problem solving
L.L ⇒ Perceiving the physical form of info.

b) Mental model :- (Experience)

→ Mental model is internal representation of a person's current understanding

→ M.M gradually developed in order to understand something, explain thing, make decision.

c) Learning :- Process of encoding in long-term memory info. from ~~to~~ short-term memory

→ Require some effort.

→ Learning can be enhanced;
- → Allow skill
- → Provides feedback.

d) Skill :- Performance is to perform skillfully

→ consistency & effort.

→ Many basic skill integrated to form increase complex one

e) Individual differences :- In a keyboard some type faster than other.

* Human consideration :-

1) The user's Knowledge & exp.

a) computer literacy b) System exp.

c) Application exp. d) Task exp.

e) other system use f) Education.

g) Reading level h) Typing skill

2) The user's task & need.

a) Type of system use b) frequency of use

c) Task or need importance d) Task structure

e) Social interaction f) Primary training

g) Turnover rate h) Job category

i) Lifestyle.

3) Psychological characteristics :-

a) Attitude b) Motivation

c) Patience d) Expectations

e) Stress level f) cognitive style.

↳ Process of gaine Knowledge from ideas.

4) The User physical characteriptical.

a) Age b) Gender c) Handedness

d) Dipability.

* Human interaction speed :-

Studies conducted by Researcher found.

1) Reading :- Text on paper (200 - 180 words/min)

2) Listening :- (150 - 160 w/m) Recommend for audio book.

3) Speaking :- Dictating to a computer (105 w/m)

4) Keying :- Typing speed (60-70 w/m)

5) Hand printing :- Human can write (Text 22 w/m)

* Method to understand User :-

1) Visit user location

2) Talk with user about their prblm.

3) Observe user working.

4) Learn about the work organization.

5) Try the job yourself.

6) Prepare surveys.

7) Research prepare charts.

* Business & Requirement Analysis :-

→ Product is developed based on input from User / marketing.

→ Many ways to capture info. for determining requirements.

1) Direct method :-

a) Individual face-to-face.

b) Telephone interview or survey

c) focus group :- A small grp of users and a moderator to verbally discuss the req.

d) facilitate team workshop

e) Observational field study.

f) card sorting for website.

2) Indirect method :- Someone b/w the user.

a) Paper survey or questionnair

b) Marketing & sales.

c) E-mail or bulletin board

d) competitor analysis.

e) System testing → feedback are obtained from ongoing product testing.

* Explain focus group :- focus group small group of user.

→ (8 to 12) & a moderator brought together to discuss the requirements.

→ User's experience, attitudes, beliefs & desires.

→ Various steps:-

a) Establish the objective of session

b) Select participants (user).

c) write script for moderator.

↳ Relevant topic

→ Ensure that everyone participate.

d) Take a good notes, using session recording.

* Card sorting for website :- Pontential contents topics are placed on individual index set card and user ask to sort the card into grouping.

→ card sorting assists in building the sit's structure.

→ Number the card on back

→ Make a record of the grouping using the number.

* Step for creating electronic survey :-

→ A questionaire or survey is administered to a sample of users via e-mail or web.

→ Speed of return is much faster than distributed in paper format.

→ In creating electronic survey.

- → Keep it short
- → ~~Kepp~~ Keep it simple.

* Basic business functions :-

→ Description of what product will do.

→ Major system function are listed & described.

→ ~~fo~~ Flowchart of major function is developed.

→ 1) Complete understanding of user's mental model

2) Performing a task analysis.

3) Develop conceptual model

↳ Describe how interface will present object & relationship b/w object.

Guidelines for designing conceptual model

- → Reflect user's mental model, habit, routine
- → Provide proper feedback
- → Provide design consistency

4) Defining objects.

5) Developing Metaphor :- use known thing to understand other similar thing.

* Guideline for Designing conceptual model

1) Present Metaphore

2) comply with expectation, habits, routines.
↳ use familier associations, avoiding the new & unfamilier.

3) Avoid unnecessary or irrelevant.

* How standards & guideline valuable to user & Developer

→ for User :- Allow faster performance
- Reduce error
- Reduce training time
- Better system utilization
- Improve satisfact

→ For Developers :- - Increase visibility of HCI
- Simplify design
- Reduce redundant effort
- Reduce training time

Thank you

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Md. Harip